

Chapter 2: Arithmetic Expressions

Worksheet 2: Arithmetic Expressions

Learning Objective: Form and evaluate arithmetic expressions correctly using brackets and the order of operations (BODMAS), and apply this to real-life billing situations.

Worked Example:

Evaluate: $12 + 3 \times (8 - 5)$.

Solution: Brackets first: $8 - 5 = 3$. Then multiplication: $3 \times 3 = 9$. Then addition: $12 + 9 = 21$.

Questions (1–20)

1. Evaluate: $6 + 4 \times 2$.

2. Evaluate: $(6 + 4) \times 2$.

3. Evaluate: $20 - 6 \div 2$.

4. Evaluate: $15 \div (5 - 2)$.

5. In $8 + 2 \times (3 + 1)$, which operation should be performed first?

6. Evaluate: $5 \times (6 + 4) - 10$.

7. Evaluate: $(12 - 4) \div 2 + 3$.

8. Insert brackets to make this true: $2 + 3 \times 4 = 20$.

9. Evaluate: $18 \div 3 + 2 \times 5$.

10. Evaluate: $100 - (20 + 30) \div 5$.

11. A shop charges Rs. 50 for entry and Rs. 20 per game played. Write an expression for the total cost of 4 games, and evaluate it.

12. Evaluate: $7 \times (3 + 2) - (10 - 4) \div 2$.

13. Find the value of: $[(8 + 2) \times 3] - 5 \times 4$.

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14. Write an expression for “twice the sum of 8 and 5, minus 6”, and evaluate it.
15. Compare using $>$, $<$ or $=$: $3 + 4 \times 2$ _____ $(3 + 4) \times 2$.
16. Using the digits 2, 3 and 4 and any operations or brackets, create an expression equal to 20.
17. A taxi charges a base fare of Rs.30 plus Rs.12 per km. Write an expression for the cost of a 7 km ride plus a Rs.15 night charge, and evaluate it.
18. Find the error in this evaluation and correct it: “ $15 - 3 \times 2 = 24$ ”.
19. A vendor sells apples for Rs.40/kg and bananas for Rs.20/kg. Write an expression for the cost of 3 kg apples and 2 kg bananas, and evaluate it.
20. For a family outing, entry tickets cost Rs.100 each for 2 adults and Rs.60 each for 3 children, plus a Rs.50 parking charge. Write and evaluate an expression for the total

cost.

Reflection Question: Why do you think mathematicians agreed on a fixed order of operations (BODMAS)? What would happen if everyone used a different order?

Real-Life Application Question: Look at a shopping bill or restaurant bill at home. Write one arithmetic expression (using the prices and quantities) that represents the total bill.